

Best Inventions of 2021

TIME

A SAVVIER WAY TO COMMUTE



A DIY ROBOTIC ARM

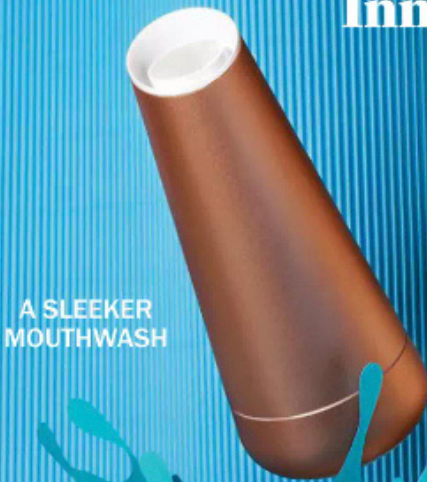


A MORE ACCURATE WORLD MAP

100 Amazing Innovations Changing How We Live



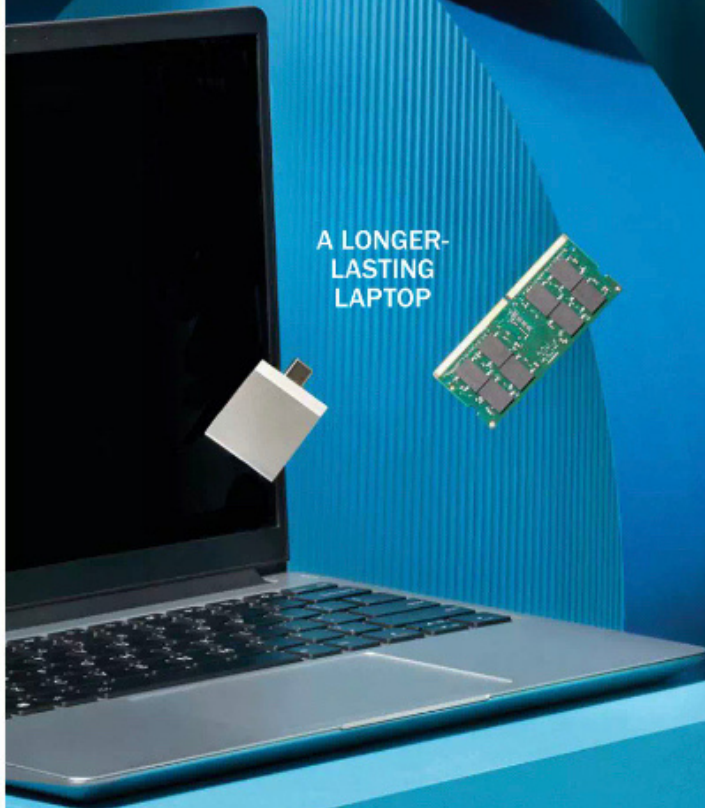
A SMARTER SLIP-ON RUNNING SHOE



A SLEEKER MOUTHWASH



A SIMPLER WAY TO FOCUS



A LONGER-LASTING LAPTOP



A TOY THAT KEEPS KIDS CALM



A MINI-CAR THAT TEACHES CODING



NEXT-GEN NOODLES

time.com

just READ
A MAGAZINE

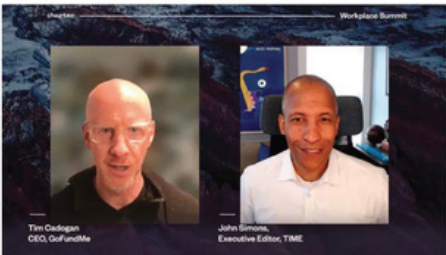
TAP HERE TO LOOK INSIDE





What you said about...

THE CLIMATE ISSUE
After reading about what's at stake for the world in the Nov. 8/Nov. 15 issue, readers shared their hopes that leaders will take bold steps to address climate change—and fears that they will fall short. “We are currently caught in a Catch-22 created by decades of pursuing economic growth,” wrote Barbara Williams of Long Hanborough, U.K. “If we fail to change our direction, we are doomed to a very messy collapse.” Readers were inspired by the young people at the forefront of the climate-change movement. “Maybe it’s time for the world’s leaders to listen to even one young voice,” wrote Dennis Fitzgerald of Box Hill, Australia. “The ones that are going to be living in the world ... long after they are gone.” Donna Snow of Lacey, Wash., was wowed by 24-year-old activist Vanessa Nakate, who appeared on the issue’s cover. “Having a young woman leading the awareness about climate change is critical for our planet,” Snow wrote. Offering “much appreciation to all those dedicated to dealing with the consequences of our disrespect and abuse of our planet,” Nanda Wilson of Felton, Calif., asked simply, “When will the U.S. give Nature a seat at the table?”



Future of work

At Charter’s virtual Workplace Summit on Nov. 9–10, TIME executive editor John Simons interviewed GoFundMe CEO Tim Cadogan and LinkedIn CEO Ryan Roslansky on the most pressing issues facing today’s business leaders, from the talent reshuffle to companies’ social impact. See more at time.com/workplacesummit



TIME in space

On Nov. 6, a copy of TIME’s Aug. 23/Aug. 30 issue featuring the crew members of Inspiration4, the first all-civilian flight into earth’s orbit, sold for \$40,000 at a charity auction for St. Jude Children’s Research Hospital. Autographed by the crew, the issue was one of several pieces of memorabilia put up for sale after going into space with the astronauts in September. As of Nov. 9, the mission has raised \$240 million for the hospital.



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On the covers



Photographs by Stacy Kranitz for TIME



Photographs by Margeaux Walter for TIME

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A NOTE TO READERS

In this issue, we are excited to begin rolling out a number of changes to the look of TIME—including a strong new type style, a fresh visual design for the table of contents, some new formats for the news analysis you’ve told us you love and new columns by TIME writers across Climate, Leadership, Health and more. We hope you like the upgrades and welcome your feedback at letters@time.com

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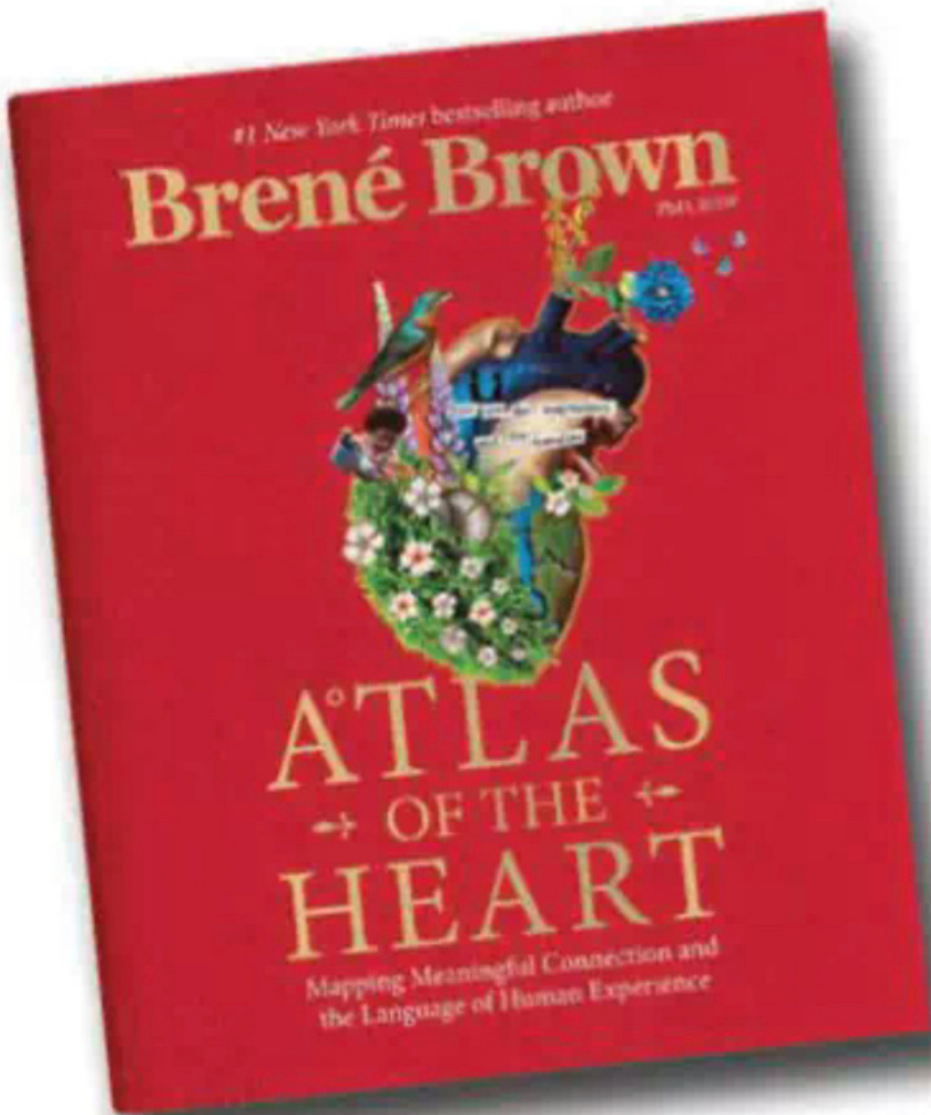
TALK TO US

Letters should include the writer’s full name, address and home telephone, and may be edited for purposes of clarity and space

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PHOTO OF WOMAN: COURTESY CLARENCE



Brown’s latest book attempts to explain 87 human emotions, why they matter and how to respond to them

TIME OFF REVIEWS

A user’s guide to human emotions

BY BELINDA LUSCOMBE

WHY IS THAT PEOPLE ARE QUICK TO SAY THEY’RE jealous of someone, but will not admit to being envious? What’s the difference between shame and guilt? Is hopelessness the same as despair? These are the questions Brené Brown, the sociology professor turned best-selling author and leadership consultant, tries to answer in her new book, *Atlas of the Heart*, out Nov. 30.

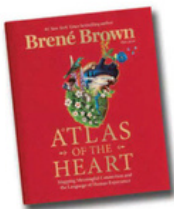
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In surveys taken by 7,000 people over five years, Brown and her team found that on average people can identify only three emotions as they are actually feeling them: happiness, sadness and anger. For Brown, who made her name by illuminating the finer contours of humans’ emotional landscape, this is not nearly enough. So she sets out to map 87 different emotions and experiences, pointing out the distinguishing features of each.

The difference between guilt and shame—for example, as fans of Brown already know from her wildly popular TED talks, her five previous No. 1 best sellers, her podcasts and Netflix lecture—is that guilt tells people they did something bad while shame tells people they are something bad. The difference between envy and jealousy is that envy materializes when one wants something somebody else has—looks, status and wealth are the big trio—while jealousy is the feeling that a relationship is being threatened. Hopelessness is a fleeting fear that a task is too difficult, and despair is a feeling that life is too difficult. And so on.

AS A POTENTIAL CARTOGRAPHER for the human experience, Brown is a solid candidate. She’s the Dr. Fauci of feelings; she can take complex subjects that require years of study and explain them in a comprehensible and reassuring way. But unlike America’s most famous public-health official, Brown is allowed to be vulnerable. Indeed, she insists on it. *Atlas* flicks at her personal biography as a source of exposition. She mentions in passing that she came from a dysfunctional but high-performing family, and that she’s a recovering alcoholic, a committed swimmer and a perfectionist.

This formula, very human subject + rigorous



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The pleasure of reading Brown’s previous work has been dawning self-recognition, as she carefully and diligently loosened the knots that bound readers’ hearts and hindered their ability to cope with their circumstances. This is more a book of quick hits, a reference guide to dip into when readers need a little reorientation. Brown’s attempt to help people find their way to one another is laudable, but she has compiled an atlas when we need a GPS.

BROWN: CHRIS PIZZELLO/AP

We look forward to seeing you in the air again

An elevated experience awaits on board
the world's most awarded airline



**SINGAPORE
AIRLINES**



A STAR ALLIANCE MEMBER



REOPENED

U.S. borders

An end to a long COVID-19 closure

IN MARCH 2020, AMERICAN skies went strangely quiet. In an effort to control the pandemic, the U.S. banned travelers from 33 countries. Flights from Europe, China, Brazil, India, South Africa and more were canceled, alongside land crossings from Canada and Mexico.



Loved ones reunited at Newark airport on Nov. 8

The View

HEALTH

HOW TO KEEP KIDS SAFE

BY MONICA GANDHI AND JEFF VERGALES

With the Pfizer-BioNTech COVID-19 vaccine newly given emergency-use authorization for children 5 to 11, many parents are asking the question: Should we vaccinate our kids? Many of them may have already made up their mind. Some of those arguing against vaccinating kids make the case that COVID-19 typically does not cause severe illness among children with healthy immune systems. ▶

INSIDE

VOTING AND VIOLENCE
IN BAGHDAD

FACE THE THING YOU
ARE MOST AFRAID OF

THE INFRASTRUCTURE BILL:
NOT SUCH A HUGE VICTORY

THE 1000 BEST INVENTIONS OF 2021

Every year, TIME highlights inventions that are making the world better, smarter and even a bit more fun.

To assemble our 2021 list, we solicited nominations from TIME's editors and correspondents around the world, as well as through an online application process. We then **evaluated** *oceniać* each contender on a number of key factors, including originality, creativity, **efficacy**, ambition and impact. *skuteczność*

The result is a list that includes an innovative, eco-friendly dye for jeans, a truly novel pasta shape, **groundbreaking** vaccines for COVID-19 and malaria—and much more.

przełomowy

PHOTOGRAPHS BY MARGE AUX WALTER FOR TIME

With reporting by Eloise Barry, Leslie Dickstein, Mariah Espada, Alejandro de la Garza, Nik Popli, Simmone Shah and Julia Zorthian

SKIN CARE ON DEMAND

Opulus Beauty Labs Treatment System

Dr. Robb Akridge, who invented the Clarisonic face-cleansing brush 17 years ago, found inspiration for his latest beauty creation in a chocolate shop in Paris.

“Most of the chocolates were individual morsels,” he says, “and each created an experience that could only be achieved at the time you bit into them.” Why couldn’t skin care be the same, he wondered. Now it is. Think of the Opulus Beauty Labs Treatment System (\$395) as a Nespresso machine for skin care—minus the plastic waste. Customers place a small paper pod, which includes two formulations, into the water-bottle-size device, which heats up to mix the formula and deliver a single-use dose of skin-care product, to be applied immediately when it is warm and more easily absorbed by the skin. Opulus currently offers a retinol treatment, and with a recent \$8 million infusion of cash from investors, more treatments are on the way.

—Taylor Bryant

na żądanie

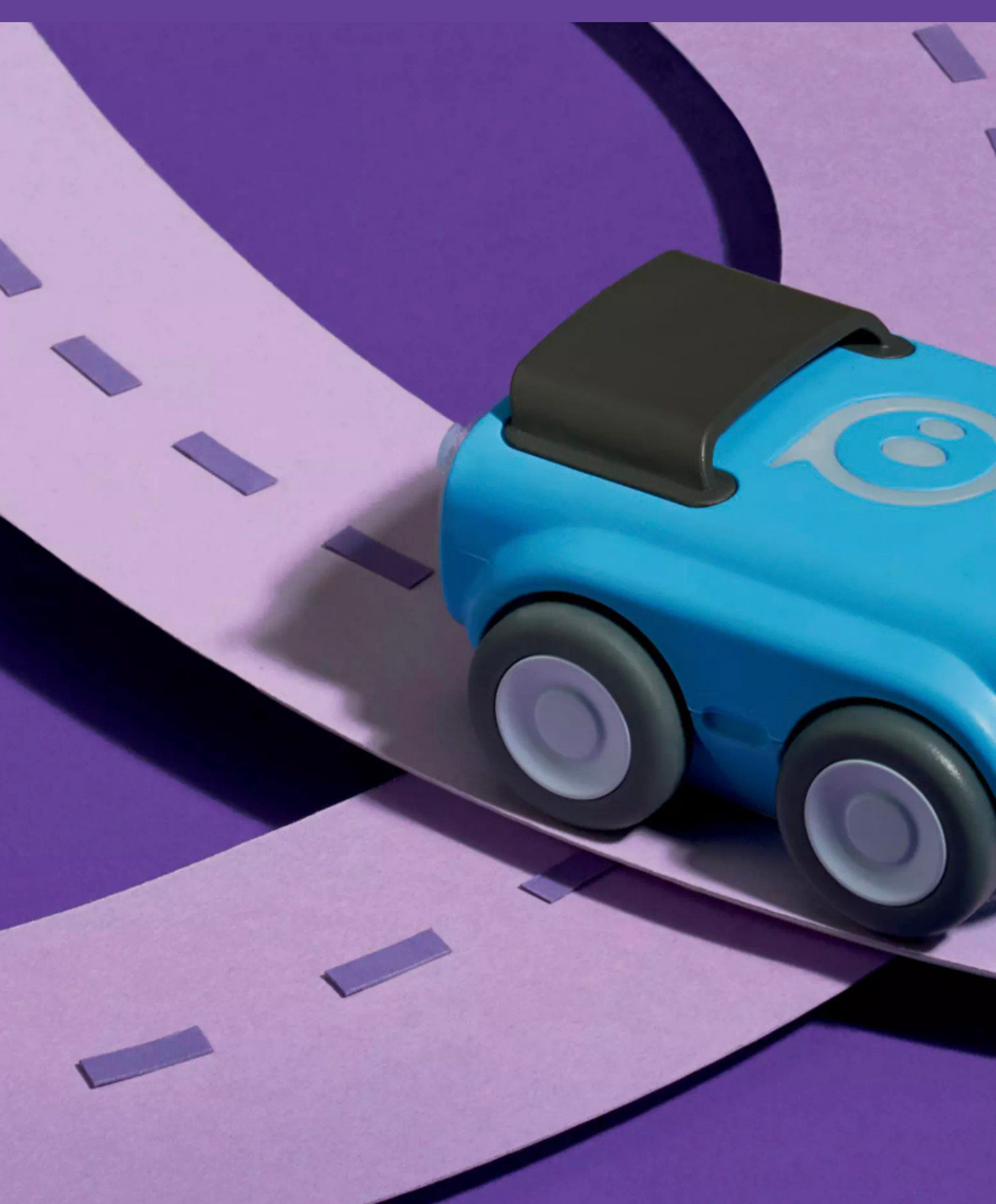
kąski



papierowe opakowanie

nagrzewać się

kuracja



EDUCATION

TEACHING KIDS TO CODE

Sphero indi

kwestia rozwiązywania problemów

At its most basic level, coding is a **matter of problem solving** and pattern recognition. Sphero indi teaches both—no screen required. Instead, kids as young as 4 use different-colored silicone tiles to move a robotic car from point A to point B—green tiles increasing its speed, pink tiles telling it to turn 90 degrees left, purple tiles telling it to spin and dance, and so on. “We really wanted to take these abstract concepts and bring them into the physical world,” says Jeff Wiencrot, one of the principal engineers who worked on the product. A student kit priced at \$124.99 includes the car and 20 colored tiles to help kids make programming-based puzzles. —Chad de Guzman



MEDICAL CARE

A PUBLIC-HEALTH BREAKTHROUGH

Mosquirix malaria vaccine

plaga

Malaria has long been a global scourge, killing some 500,000 people a year, more than half of them children under 5. But that may be about to change, thanks to the introduction of a malaria vaccine—the first vaccine ever against a parasitic infection. The shot has been in the works since 1987, at a cost of more than \$750 million, mostly funded by the Bill and Melinda Gates Foundation and GlaxoSmithKline. Dubbed Mosquirix, it was recommended for approval by the World Health Organization in October, after a field trial involving 800,000 children in Ghana, Kenya and Malawi. The four-dose regimen was found to cut the risk of infection by 40% and the risk of severe infection by 30%. —Jeffrey Kluger



FOOD & DRINK

Catch of tomorrow

Kuleana tuna

oparty na roślinach

Tasty and nutritious **plant-based** alternatives for meat and chicken have been available for years. But seafood? Not so much. That's the void that Kuleana is trying to fill with its 100% plant-based, sushi-grade, ready-to-eat tuna made from ingredients including algae, koji (a fungus that grows in East Asia), radish, bamboo and potato. Deep red in color and designed to be prepared as sushi, *nigiri*, carpaccio, poke or ceviche, the alt-tuna retains the iron, vitamin B12 and omega-3 fatty acid of the real thing—without the microplastics, mercury or high cholesterol. And the benefits are more than nutritional—it may also help to **alleviate** reliance on industrial fishing in the face of increasing demand for fresh food. The product is currently available at select markets in Los Angeles, restaurants in the Midwest and Poké Bar locations **nationwide**, with a wider rollout via e-commerce slated for the near future. Next on the menu: high-quality, sushi-grade, plant-based salmon. —*Sanya Mansoor*

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złagodzić

AR & VR

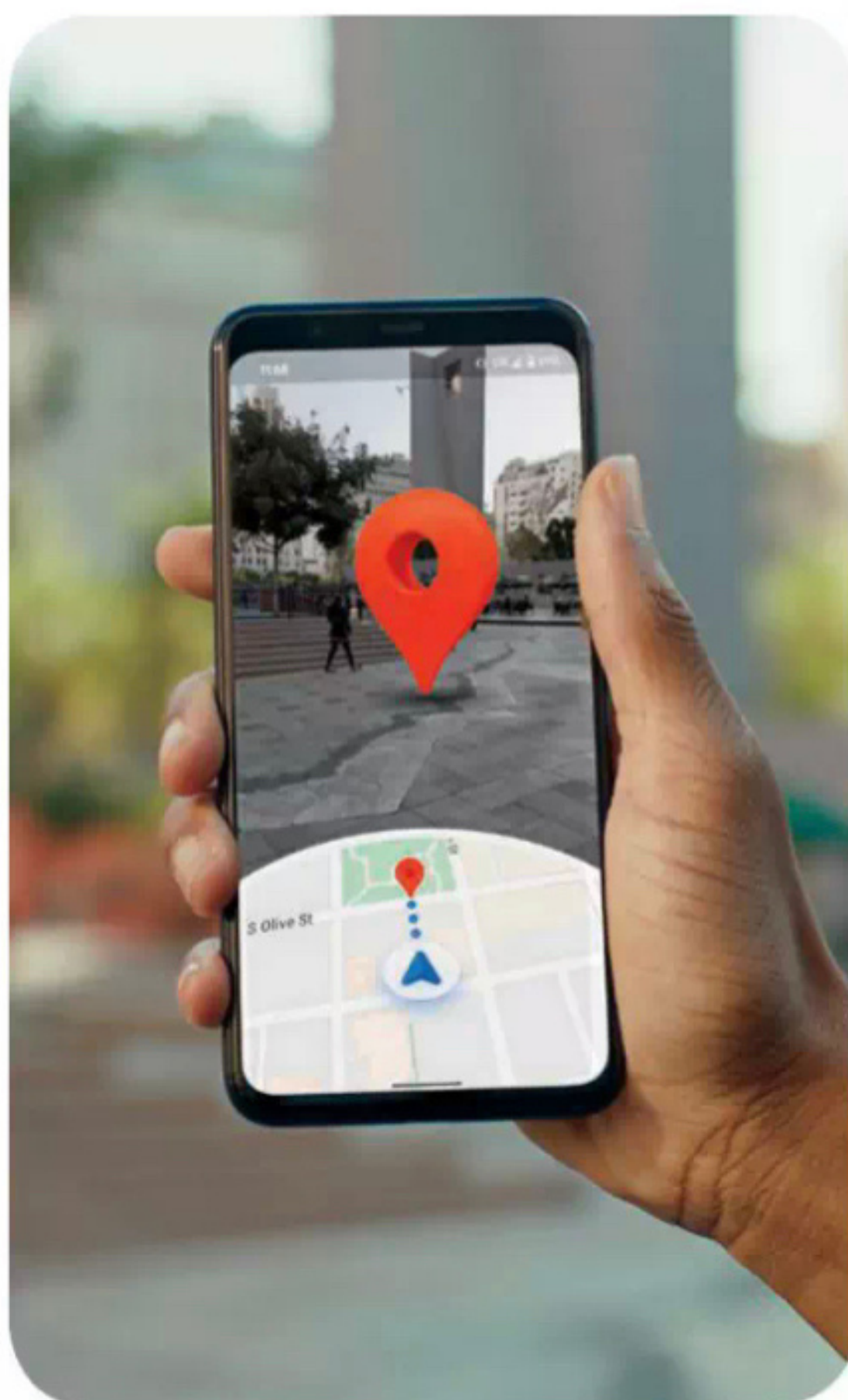
LOOK WHERE YOU'RE GOING

Google Maps Live View

How many times have you wandered around an unknown location, staring at the map on your phone and wondering where to turn? As a pedestrian, orienting oneself with bird's-eye directions is far from intuitive. Google Maps' free Live View mode fixes that, using augmented

reality to lay instructions on top of what you're seeing live on the street via your smartphone camera's viewfinder—"effectively aligning your map with the real world," says Andre Le, lead designer for Live View. So far, Google has Street View coverage for about half the world, and Live View was launched in tricky multi-level transit hubs, airports and malls in Tokyo prior to the Olympic Games (as well as in Zurich). The company is still piloting indoor locations in the U.S., such as shopping malls, but tens of millions have already used Live View globally.

—Alison Van Houten





ACCESSIBILITY

Mobility for the littlest kids

Permobil Explorer Mini

Children who are slow to walk because of **diseases** like muscular dystrophy, cerebral **palsy** or Down syndrome have been shown to benefit **cognitively** and socially by moving around with the help of a motor. The problem: most motorized wheelchairs are too big for the tiniest kids. As a result, some parents of such children have been stuck modifying toy cars to help them out. Enter Permobil's Explorer Mini (\$2,695): a U.S. Food and Drug Administration–cleared motorized device built expressly for kids ages 1 to 3. Kids can control the chair, available only with a prescription, via a joystick. The idea, says Amy Morgan, a pediatric physical therapist and Permobil project manager, is “to jump-start [kids'] physical development.” —J.W.

choroby

porażenie

poznawczo



FOOD & DRINK

A NOVEL NOODLE

Sfoglioni Cascatelli

sosowatość

łatwo

No pasta shape was good enough for Dan Pashman. The obsessive foodie and host of the *Sporkful* podcast judges pasta shapes on sauceability (how readily sauce adheres), forkability (ease of getting and keeping it on a fork) and toothsinkability (how satisfying it is to sink your teeth into). “Most of the pasta shapes out there are only good at one or two,” he says. So Pashman set out to create his dream pasta shape, an experience chronicled on *The Sporkful*. Three years and several prototypes later, he finally unveiled Cascatelli, with a half-tube shape to catch sauce and ruffles to provide a dynamic bite, all in a short noodle that’s easily forked. Now manufactured by Sfoglioni (\$19.96 for 4 lb.), it’s sold nearly 300,000 lb. since debuting in March. —*Eliana Dockterman*

BOOKS

A user's guide to human emotions

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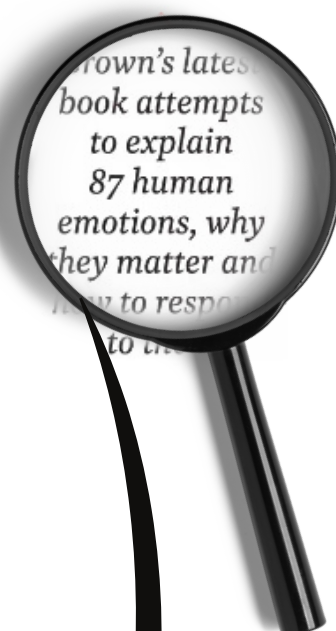
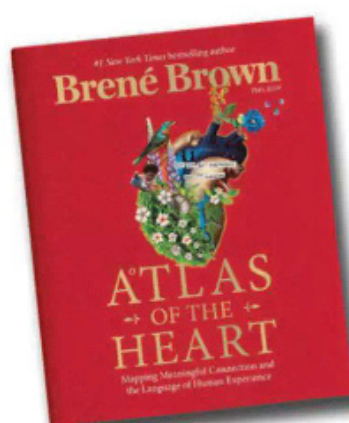
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This formula, very human subject + rigorous

80 TIME November 22/November 29, 2021

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TIME 2030 at COP26

AS NEGOTIATIONS AMONG WORLD leaders continued at COP26, the 2021 U.N. climate-change conference in Glasgow, Scotland, TIME convened its own solution-focused roundtable.

skoncentrowany na rozwiązaniach

On Nov. 2, TIME hosted an event for more than 100 conference attendees. The evening included a keynote panel moderated by TIME co-owner and Salesforce chair and CEO Marc Benioff and featured U.S. special presidential envoy for climate John Kerry; former Vice President Al Gore, a Nobel Peace Prize winner for his work on climate change; Ugandan climate activist Vanessa Nakate, founder of the Rise Up movement; and designer Stella McCartney, a sustainable fashion pioneer. The event, part of TIME's decade-long 2030 initiative focused on how to build a healthier, more resilient planet, was held in partnership with SOMPO Holdings and BCG.

uczestnicy

tutaj: poprzedni

zrównoważony

odporny



The Brief



The world's largest carbon emitters now all have net-zero pledges. Most of them are

zobowiązania

vague

niejasny

GOOD QUESTION

Have U.S. shoppers broken the supply chain?

łańcuch dostaw

AMERICA HAS LONG GOBBLED UP MORE **pochłonięty** goods from overseas than it sends back, but in the past year, spending has gone **bonkers**. **zwariowany** Americans imported \$240 billion worth of goods in September, up 16% from September 2019. Meanwhile, the country exported about \$81 billion less than it imported—a record trade deficit, according to data released by the U.S. Census Bureau on Nov. 4. Spending will likely continue **apace** as consumers **szybko** stock up for the holidays.

People largely stopped shopping in March 2020 as they hunkered down amid the early months of the COVID-19 pandemic, so shipping companies deployed containers for other tasks. When e-commerce spending started booming that summer, shipping containers were **stranded** **na mieliźnie** in ports across the world. As backlogs grew, factories everywhere could no longer rely on the global supply chain to get them the parts they needed quickly, which meant delays in getting products to consumers. Meanwhile, big **retailers** **detałiści** rushed to load up warehouses to try and get goods to customers soon after they were ordered online.



“Americans have become singularly impatient consumers,” says Jock O’Connell, an international trade adviser at Beacon Economics. “We have much more inventory in this country than we would need if Jeff Bezos hadn’t convinced us we could have our stuff delivered tomorrow.”

Because there are so many more goods coming into the nation, U.S. ports are **clogged**. Consumer spending is usually good for the economy, but the bottlenecks created by America’s huge appetite are becoming a problem for economic growth. The U.S. economy grew at a rate of just 2% from July to September, the Commerce Department said on Oct. 28, down significantly from the 6.7% it grew the quarter before, largely because of supply-chain **bottlenecks**. “We’re in the midst of unprecedented cargo congestion,” said Gene Seroka, executive director of the Port of Los Angeles, on Oct. 29. “The system is all **gummed up**.”

zatkany

wąskie gardła

zaklezione



DIED

Astroworld concertgoers

Music festival tragedy

EIGHT PEOPLE DIED AND HUNDREDS MORE WERE INJURED AT the Astroworld Festival in Houston on Nov. 5, when a crowd of some 50,000 people, eager to see the rapper Travis Scott, pressed relentlessly toward the main stage. Scott prides himself on the ferocity of his live shows, and has been arrested twice for encouraging his fans to storm his stages. On Nov. 5, many in the audience surged violently upon his arrival, causing some people to fall. This in turn resulted in additional panic; people began passing out in the middle of the scrum, where it was hard for them to escape or receive aid. Concert medics have also said they were dangerously understaffed.

The show continued for at least half an hour after officials declared a “mass-casualty event.” Videos of concertgoers screaming, “Stop the show!” to no avail circulated widely on social media; other clips showed individuals being crowdsurfed to safety. (Scott did pause his set at least once to direct security to help unconscious people.) Extremely young concertgoers were among the dead, including two teenagers; the six other victims were confirmed as people in their 20s. At least 18 civil lawsuits have been already filed seeking damages for death or injuries, with Scott, the rapper Drake—who performed onstage alongside him—and Live Nation Entertainment Inc., the festival’s promoter, among those named as defendants. Injury lawyers are seeking settlements in the millions of dollars. Local authorities, meanwhile, have begun an investigation into whether the festival’s organizers were criminally negligent. —ANDREW R. CHOW

relentlessly - **bezlitośnie**

ferocity - **okrucieństwo**

encouraging - **zachęcający**

passing out - **omdlenie**

understaffed - **mieć za mały personel**

mass-casualty event - **impreza masowa**

unconscious - **nieświadomy**

defendants - **pozwani**